

GRK SERVICES TRUCKING MANAGEMENT SYSTEM

UGRD-ITE6300 Cloud Computing and Internet of Things

UGRD-IT6314 Web Application Development 1

Bachelor of Science in Information Systems

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1. Introduction

"Here and Now" consumption philosophy of Filipino consumers The industry is experiencing a massive surge. The Philippine logistics market reached \$31.1 Billion in 2025 and is projected to grow steadily through 2034 (IMARC Group, 2026). Logistic tracking in the Philippines evolved from manual, paper-based to transitioning into a modern, tech-driven system fueled by e-commerce. Specialized logistics companies, such as FAST Logistics, went beyond basic freight forwarding in the 2000s and entered organized warehousing, cold chain management, and distribution. E-commerce growth. There is a huge demand for last-mile delivery due to the increased use of smartphones and the internet. The use of automated, end-to-end cargo tracking systems was expedited by the COVID-19 epidemic. GRK Trucking Service is a tracking service. First name "Georizz Hauling Services" to 1996,2017 "Grk trucking services"" Located at Capulong St., Quezonville Tayuman Ext., Tondo Manila. tracking service in Metro Manila, exclusively relying on manual process for booking, clients will book through calls, and/or. This often results in inefficiencies like poor management, delayed services, and unorganized

1.1. Background of the Study

This document serves as a comprehensive technical proposal and comparative analysis focused on the digital transformation of a local logistics firm. It outlines our group's research methodology, starting with a study of the **Lalamove ecosystem** to establish a logical blueprint for modern delivery phases: booking, active tracking, and completion. By documenting our technical stack, database architecture, and project goals, this paper establishes a formal framework for transitioning a legacy business from manual, paper-based workflows into a data-driven Information Systems environment.

The project itself centers on the development of the **GRK Trucking Services Web-based Booking and Tracking Management System**, a specialized cloud-integrated platform designed for B2B hauling operations. Utilizing a technical stack of **HTML, CSS, JavaScript, and PHP**, the system is engineered to automate core workflows and maintain data integrity through a structured **MySQL** database. By deploying the platform via **Amazon Web Services (AWS)**, we are providing the company with a scalable, centralized dashboard that allows for real-time fleet management and accessible record-keeping from any location.

The primary problem we are solving is the "manual bottleneck" caused by **GRK Trucking's** sole reliance on phone-based reservations and analog record-keeping. Currently, these legacy methods lead to disorganized data, frequent scheduling

conflicts, and significant service delays that cannot keep pace with modern automated standards. Our system specifically targets these systemic inefficiencies by eliminating human error in the booking process, reducing customer wait times, and providing the digital transparency needed for the company to remain competitive in a \$31-billion industry.

1.2. Objectives

2. **To Automate Reservation Management:** Transition from phone-based bookings to a digital interface using **PHP** and **MySQL** to eliminate data redundancy and prevent scheduling conflicts.
3. **To Implement Real-Time Status Tracking:** Develop a module that allows both administrators and customers to monitor the stages of a delivery (e.g., Pending, Picked Up, Completed), inspired by modern logistics standards like Lalamove.
4. **To Optimize Fleet and Driver Assignment:** Create a centralized dashboard for administrators to efficiently link available vehicles and drivers to specific customer bookings, ensuring clear accountability and resource utilization.
5. **To Ensure Data Integrity and security, design a robust Relational Database (ERD)** to securely store customer information, transaction history, and driver records, preventing the loss of information common in manual systems.
6. **To Leverage Cloud Scalability:** Deploy the system via **Amazon Web Services (AWS) EC2** to guarantee 24/7 accessibility and provide a platform capable of growing with the company's expanding hauling operations in Metro Manila.

7. **To Evaluate System Functionality:** Conduct rigorous testing (Unit and Functional Testing) to ensure the system meets all business requirements, as evidenced by successful test case results for reservation saving and status accuracy.

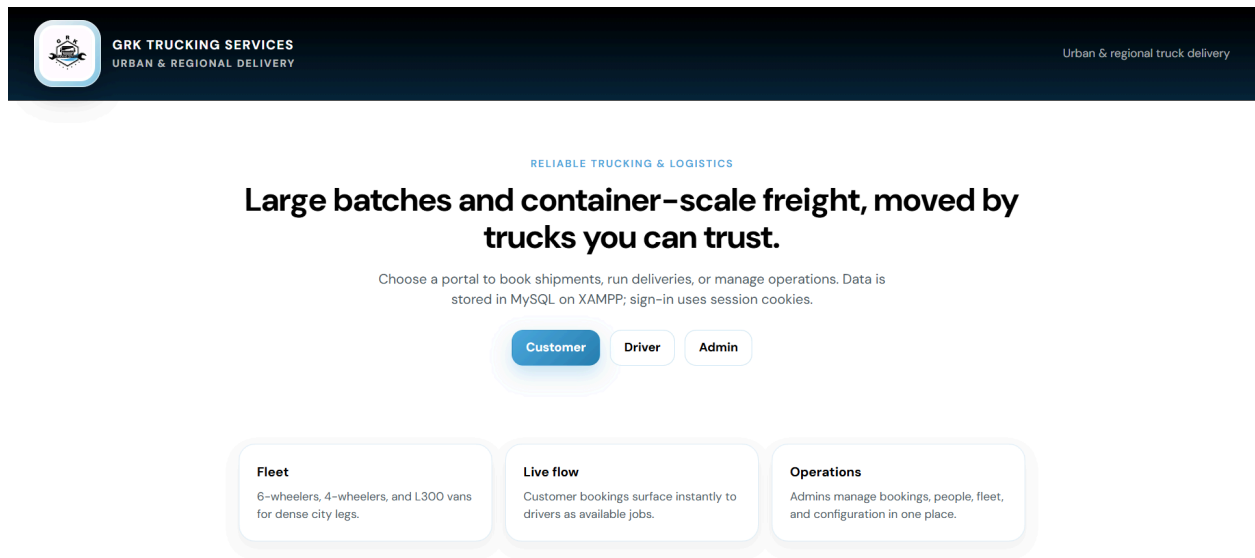
1.3 Scope

- **B2B Web-based Booking Portal:** A digital storefront where corporate clients can submit hauling requests, specify cargo details, and set pickup/drop-off points, eliminating the need for phone calls.
- **Real-Time Tracking Dashboard:** Inspired by the Lalamove model, this feature provides a live visualization of the delivery lifecycle, categorized into three distinct phases: Booking, Active Tracking, and Completion.
- **Administrative Fleet Management:** A centralized "command center" that allows the admin to monitor vehicle availability and assign specific drivers to pending customer bookings.
- **Automated Status Updates:** A system-driven workflow that enables drivers to update the progress of a delivery (e.g., "Picked Up" or "Delivered"), which is then instantly reflected on the customer's dashboard.
- **Centralized Database Management:** A robust MySQL back-end that securely stores customer profiles, driver credentials, and historical transaction data, ensuring that no records are lost or duplicated.
- **Reporting and Proof of Delivery:** The ability for the system to generate digital logs and receipts, providing both the company and the client with a permanent record of every successful transaction.

2 Features

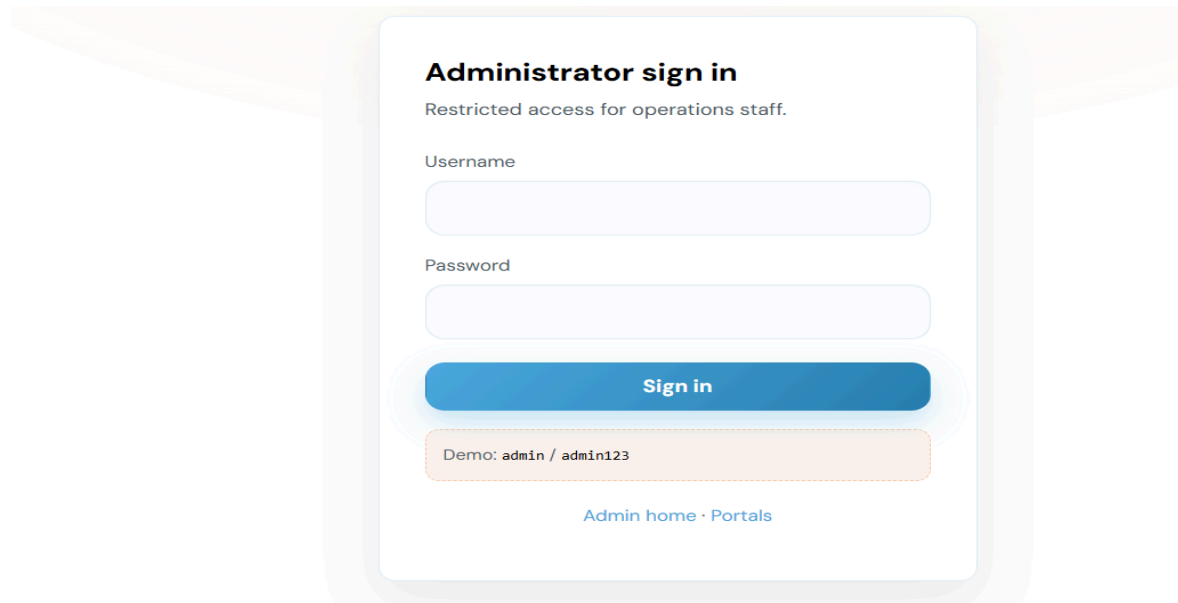
2.1 Feature 1: Landing Page

GRK Trucking Services' home portal is a platform designed for urban and regional truck delivery specializing in large batches and container-scale freight. It functions as a gateway for three distinct user types to access their respective management tools.



2.2 Feature 2: Admin Dashboard

The Administrator Sign-In page, which serves as a secure gateway for authorized personnel to access the backend or management side of a system.



Administrator sign in
Restricted access for operations staff.

Username

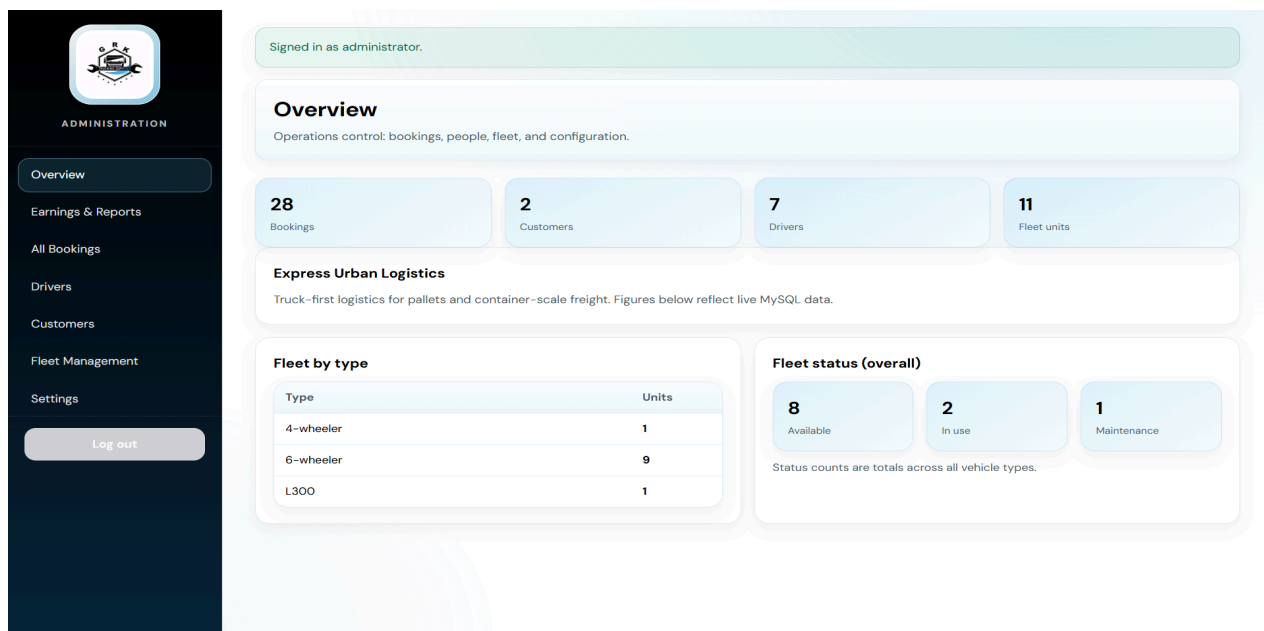
Password

Sign in

Demo: admin / admin123

[Admin home](#) · [Portals](#)

The Overview from Admin Dashboard, this tool is designed to streamline the procurement and fulfillment cycles by ensuring that logistics managers can quickly match available drivers and specific vehicle types to incoming customer bookings.



Signed in as administrator.

Overview

Operations control: bookings, people, fleet, and configuration.

28
Bookings

2
Customers

7
Drivers

11
Fleet units

Express Urban Logistics

Truck-first logistics for pallets and container-scale freight. Figures below reflect live MySQL data.

Fleet by type

Type	Units
4-wheeler	1
6-wheeler	9
L300	1

Fleet status (overall)

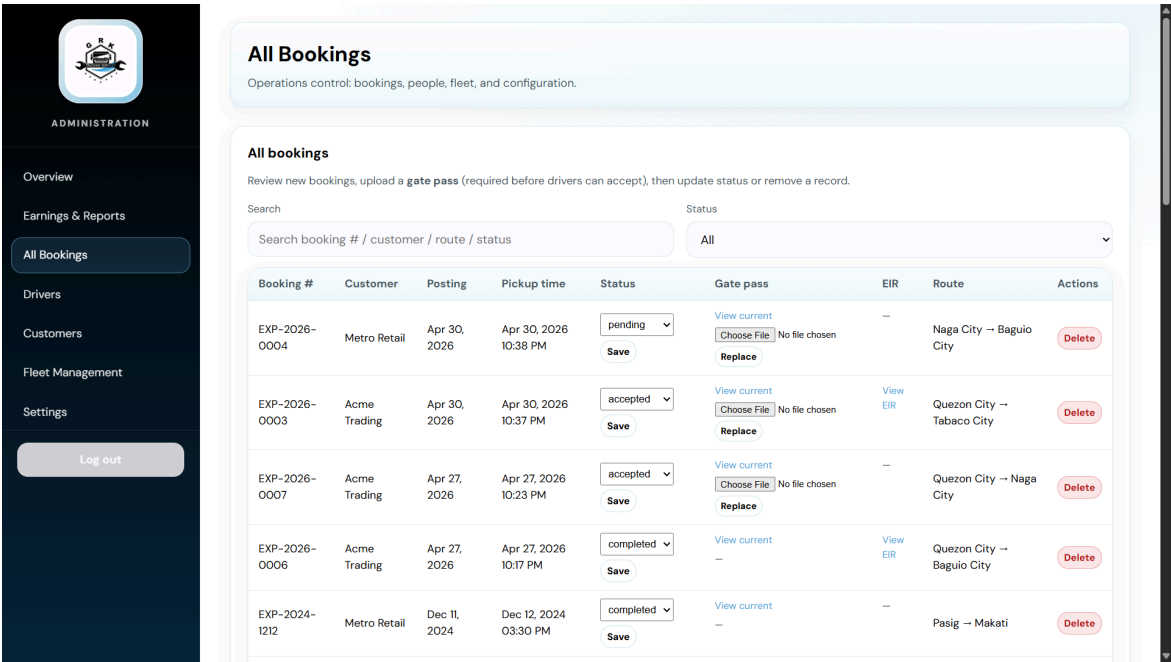
8
Available

2
In use

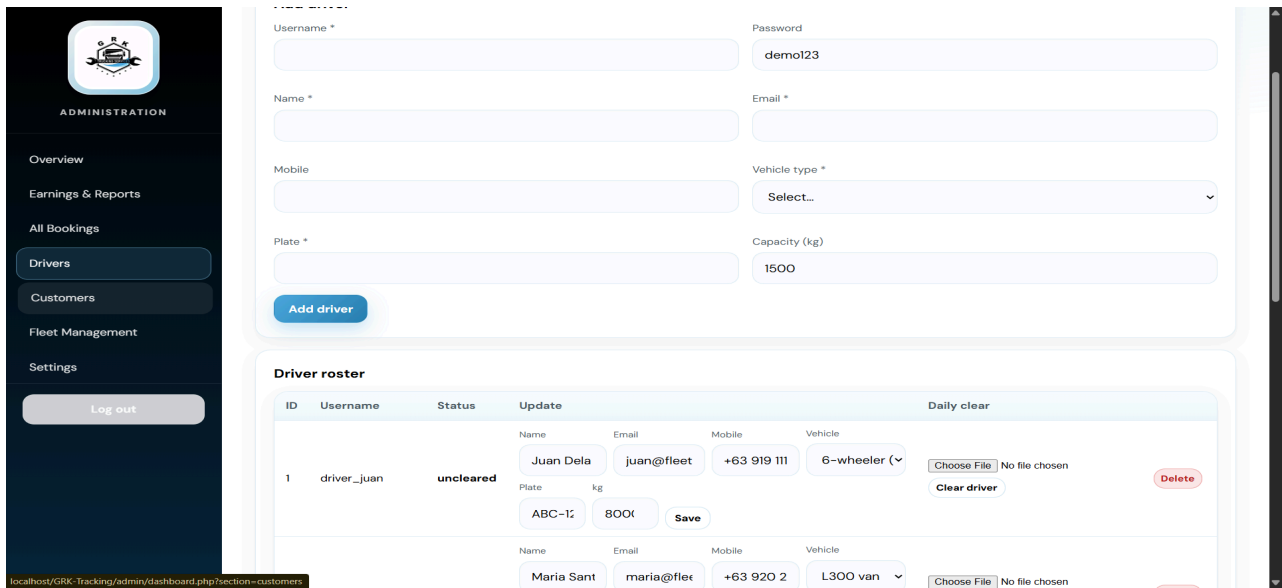
1
Maintenance

Status counts are totals across all vehicle types.

The All Bookings interface for Grk Trucking Services serves as the primary operational management screen for handling specific freight orders. It allows administrators to track the lifecycle of a booking from initial request to completion.



The Drivers section in the Administration dashboard provides high-level control over the entire workforce and fleet capacity.



The Profile section of the Driver Portal allows individual drivers to manage their professional identity and equipment specifications.



DRIVER PORTAL

Overview

Available Jobs

My Deliveries

Profile

Log out

Profile

Accept jobs, run deliveries, and keep your vehicle profile up to date.

Driver & vehicle

Full name *

Juan Dela Cruz

Username

driver_juan

Email *

juan@fleet.test

Mobile *

+63 919 111 1111

Vehicle type *

6-wheeler (Isuzu / Fuso)

Plate number *

ABC-1234

Capacity (kg)

8000

Save

The Customer list section serves as a central database for managing client relationships and account access.



ADMINISTRATION

Overview

Earnings & Reports

All Bookings

Drivers

Customers

Fleet Management

Settings

Log out

Operations control bookings, people, fleet, and configuration.

Add customer

Username *

Password

demo123

Name *

Email *

Mobile

Add customer

Customer list

Search customers

Search username / name / email / mobile

ID	Username	Update details			
		Name	Email	Mobile	
1	acme_corp	Acme Trading	orders@acme.test	+63 917 000 0001	Save Delete
2	metro_retail	Metro Retail	logistics@metro.test	+63 918 000 0002	Save Delete

The **Fleet Management** module provides the tools necessary to maintain the company's physical logistics capacity. This includes Inventory Expansion, Real-Time Fleet Dynamic, Status Updates, and Record Removal


ADMINISTRATION

Overview

Earnings & Reports

All Bookings

Drivers

Customers

Fleet Management

Settings

Log out

Label *
e.g. Isuzu F-Series

Type *
Select_

Plate *

Capacity (kg)
0

Status
available

Add to fleet

Fleet inventory

Search
Search label / plate / type / status

Status
All

Type
All

ID	Label	Update
1	Juan Dela Cruz unit	Label Juan Dela Cruz unit Type 6-wheeler Plate ABC-123 kg 800 Status in_use Save Delete
2	Maria Santos unit	Label Maria Santos unit Type L300 Plate XYZ-567 kg 150 Status available Save Delete

The **Settings** section serves as the global configuration hub for the entire platform. Its primary functions include Company Branding, Communication Channels, Regional Scope, Database Integration.


ADMINISTRATION

Overview

Earnings & Reports

All Bookings

Drivers

Customers

Fleet Management

Settings

Log out

Settings

Operations control: bookings, people, fleet, and configuration.

System configuration

Values are stored in the MySQL [settings](#) table.

Company name *
Express Urban Logistics

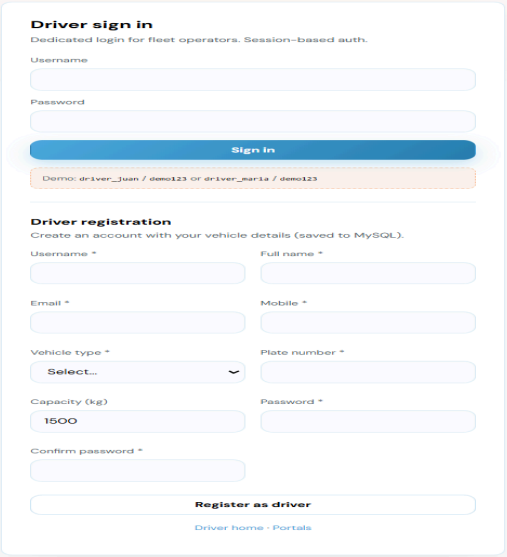
Support email *
support@express.test

Default service region
NCR + Calabarzon

Save settings

2.3 Feature 3: Driver Dashboard

The Driver Sign In portal is for existing fleet operators. It utilizes session-based authentication to manage secure access to the driver's dashboard. The Users must enter a valid Username and Password to sign in.



The image shows a digital form for driver sign-in and registration, presented on a tablet-like device. The form is divided into two main sections: 'Driver sign in' and 'Driver registration'.

Driver sign in
Dedicated login for fleet operators. Session-based auth.

Username

Password

Sign in

Demo: driver_juan / demo123 or driver_maria / demo123

Driver registration
Create an account with your vehicle details (saved to MySQL).

Username *

Full name *

Email *

Mobile *

Vehicle type *

Plate number *

Capacity (kg)

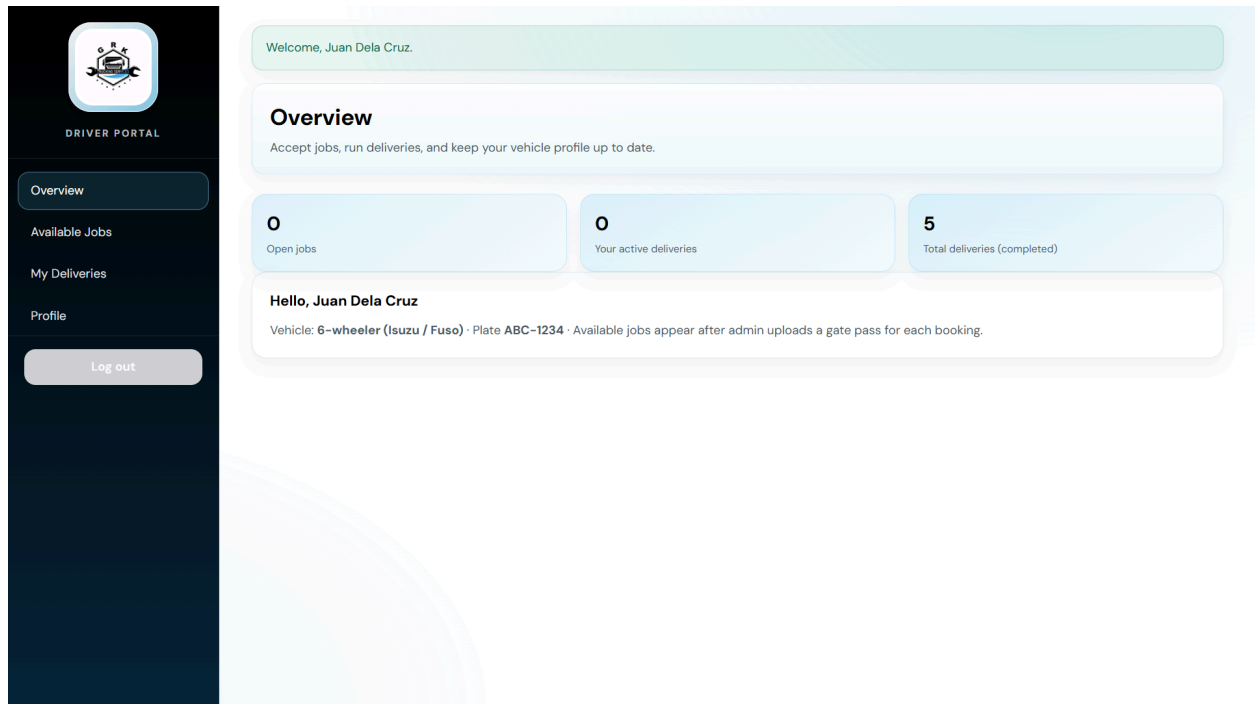
Password *

Confirm password *

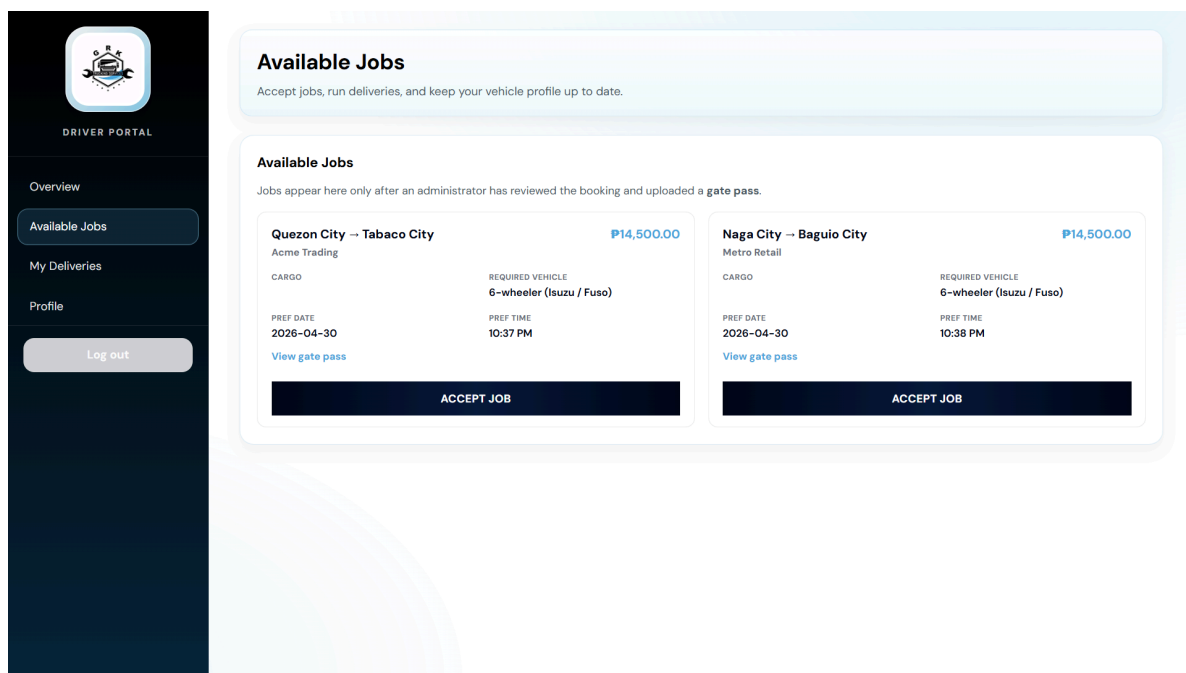
Register as driver

[Driver home](#) · [Portals](#)

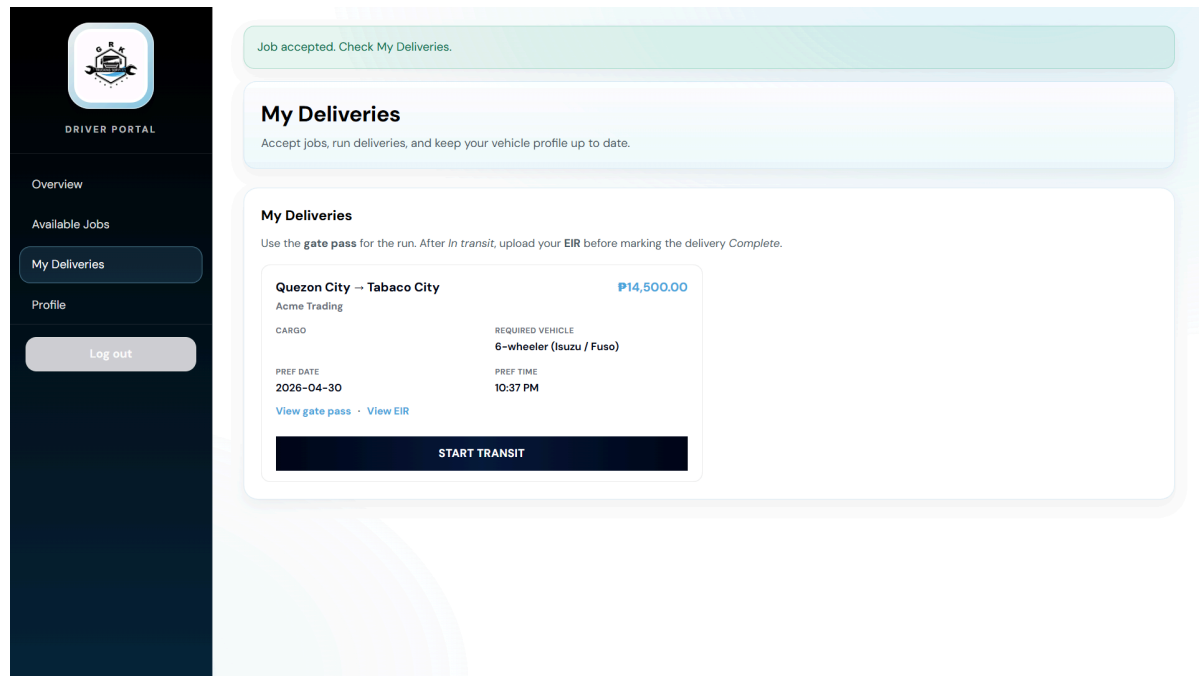
This personal dashboard provides drivers with a summary of their performance, including their total completed deliveries and current active assignments. It also displays their assigned vehicle information for easy reference.



This section acts as a job board where drivers can browse available delivery assignments. Each listing includes the route, the customer, the payout amount, and required vehicle specifications.



This operational screen allows drivers to manage their accepted tasks by initiating the "transit" phase. It also facilitates the necessary documentation workflow, such as viewing gate passes and uploading required receipts like EIR.



The Profile page enables drivers to self-manage their professional identity and equipment records. Drivers can update their contact information and ensure their vehicle's plate number and capacity are accurately reflected in the system.

DRIVER PORTAL

- Overview
- Available Jobs
- My Deliveries
- Profile**
- Log out

Profile

Accept jobs, run deliveries, and keep your vehicle profile up to date.

Driver & vehicle

Full name *

Username

Email *

Mobile *

Vehicle type *

Plate number *

Capacity (kg)

Save

8.4. Feature 4: User Dashboard

This entry page provides a separate login for booking customers using session-based authentication. It also features a registration form for new users to create an account by providing their full name, email, mobile number, and vehicle requirements to start managing shipments.

Customer sign in

Separate login for booking customers. Session-based auth.

Username

Password

Sign in

Demo: [ecm Corp](#) / [demo123](#) or [metro_retail](#) / [demo123](#)

Create an account

Register to book trucks and manage shipments.

Username *

Full name *

Email *

Mobile *

Password *

Confirm password *

Register

[Customer home - Portals](#)

This dashboard

provides a high-level summary of the user's activity, showing total bookings and the status of shipments currently "Awaiting driver" or "In progress". It includes quick actions

to start a new booking or track an existing shipment, and lists the support contact and service region (NCR + Calabarzon).

The screenshot shows the 'Overview' page of the Customer Portal. On the left is a dark sidebar with a logo at the top, followed by the text 'CUSTOMER PORTAL'. Below this are links: 'Overview' (highlighted), 'Booking', 'Track Shipment', 'My Bookings', and 'Profile'. At the bottom of the sidebar is a 'Log out' button. The main content area has a light blue header with 'Welcome back, Acme Trading.' Below this is an 'Overview' section with a description: 'Truck-based delivery for large batches and container-scale urban & regional moves.' This is followed by three summary cards: '15 Total bookings', '0 Awaiting driver', and '0 In progress'. A 'Welcome, Acme Trading' section contains a message about booking vehicle types (6-wheelers, 4-wheelers, or L300 vans) and a note about 'Available Jobs'. Below this are 'Quick actions' with buttons for 'New booking' and 'Track', and a 'Need help?' section with support contact information: 'Support: support@express.test - Region: NCR + Calabarzon'.

This page allows customers to request specific vehicle types, such as 6-wheelers, 4-wheelers, or L300 vans, for pallet and container-scale moves. Requests remain "Pending" until an administrator reviews them and uploads a gate pass, after which a driver can accept the job.

The screenshot shows the 'Truck Booking' page. The sidebar is identical to the previous page, but the 'Booking' link is now highlighted. The main content area has a light blue header with 'Truck Booking' and a description: 'Truck-based delivery for large batches and container-scale urban & regional moves.' Below this is a 'Request a truck' section. It starts with a status message: 'Your request is saved as Pending until an administrator reviews it and uploads a gate pass. Only then can a driver accept the job.' This is followed by a note: 'Availability is based on Fleet Management units marked available.' The form contains several fields: 'Full name / company *' (filled with 'Acme Trading'), 'Email *' (filled with 'orders@acme.test'), 'Mobile *' (filled with '+63 917 000 0001'), and 'Vehicle type *' (a dropdown menu with 'Select...' selected). There are also fields for 'Preferred pickup date/time *' (filled with '30 Apr 2026 10:34 pm' and a calendar icon), 'Pickup location *' (filled with 'Warehouse / city'), 'Drop-off location *' (empty), and 'Cargo description' (filled with 'Pallets, containers, dimensions...').

Customers can use this interface to check the real-time status of a delivery by entering a booking number or username. It displays detailed shipment and booking information, including contact details, pickup/drop-off routes, and the current status (e.g., "Completed").



CUSTOMER PORTAL

Overview

Booking

Track Shipment

My Bookings

Profile

Log out

Track Shipment

Truck-based delivery for large batches and container-scale urban & regional moves.

Check status

Enter a **booking number** for one shipment, or a **username** to list all bookings for that account.

Booking number

EXP-2026-0003

Username

acme_corp

Search

Shipment & booking details

Tracking reference **EXP-2026-0003** Completed

BOOKING & CONTACT

Account username	acme_corp
Contact name	Acme Trading
Email	orders@acme.test
Mobile	+63 917 000 0001
Booking date & time	Apr 27, 2026 10:50 PM
Posted	Apr 27, 2026

ROUTE

Pickup	Naga City
Drop-off	Baguio City

VEHICLE & CARGO

This page lists all historical and active bookings tied to the customer's account in a table format. It provides a comprehensive view of posting dates, pickup times, current statuses, vehicle types used, and the specific delivery routes.



CUSTOMER PORTAL

Overview

Booking

Track Shipment

My Bookings

Profile

Log out

My Bookings

Truck-based delivery for large batches and container-scale urban & regional moves.

Your shipments

All bookings tied to your account.

Booking #	Posting	Pickup time	Status	EIR	Vehicle	Route
EXP-2026-0003	Apr 27, 2026	Apr 27, 2026 10:50 PM	Completed	—	6-wheeler (Isuzu / Fuso)	Naga City → Baguio City
EXP-2026-0007	Apr 27, 2026	Apr 27, 2026 10:23 PM	Accepted	—	6-wheeler (Isuzu / Fuso)	Quezon City → Naga City
EXP-2026-0006	Apr 27, 2026	Apr 27, 2026 10:17 PM	Completed	—	6-wheeler (Isuzu / Fuso)	Quezon City → Baguio City
EXP-2024-1211	Dec 10, 2024	Dec 11, 2024 02:00 PM	Completed	—	6-wheeler (Isuzu / Fuso)	Manila → QC
EXP-2024-1209	Dec 8, 2024	Dec 9, 2024 10:30 AM	Completed	—	6-wheeler (Isuzu / Fuso)	QC → Taguig
EXP-2024-1207	Dec 6, 2024	Dec 7, 2024 08:00 AM	Completed	—	6-wheeler (Isuzu / Fuso)	Bulacan → Makati
EXP-2024-1205	Dec 4, 2024	Dec 5, 2024 12:00 PM	Completed	—	Reefer / specialized	QC → Pasig
EXP-2024-1203	Dec 2, 2024	Dec 3, 2024 10:00 AM	Completed	—	L300 van	Makati → Manila
EXP-2024-1201	Nov 30, 2024	Dec 1, 2024 08:00 AM	Completed	—	6-wheeler (Isuzu / Fuso)	QC → Makati
EXP-2024-1001	Oct 4, 2024	Oct 5, 2024 09:00 AM	Completed	—	L300 van	Makati → BGC
EXP-2024-0801	Aug 18, 2024	Aug 19, 2024 03:00 PM	Completed	—	6-wheeler (Isuzu / Fuso)	Bulacan → Manila

This section allows users to manage their personal and contact information, including their display name, email, and mobile number. It notes that the account username is fixed for login purposes.



CUSTOMER PORTAL

Overview

Booking

Track Shipment

My Bookings

Profile

Log out

Profile

Truck-based delivery for large batches and container-scale urban & regional moves.

Personal & contact

Display name *

Acme Trading

Username

acme_corp

Email *

orders@acme.test

Mobile *

+63 917 000 0001

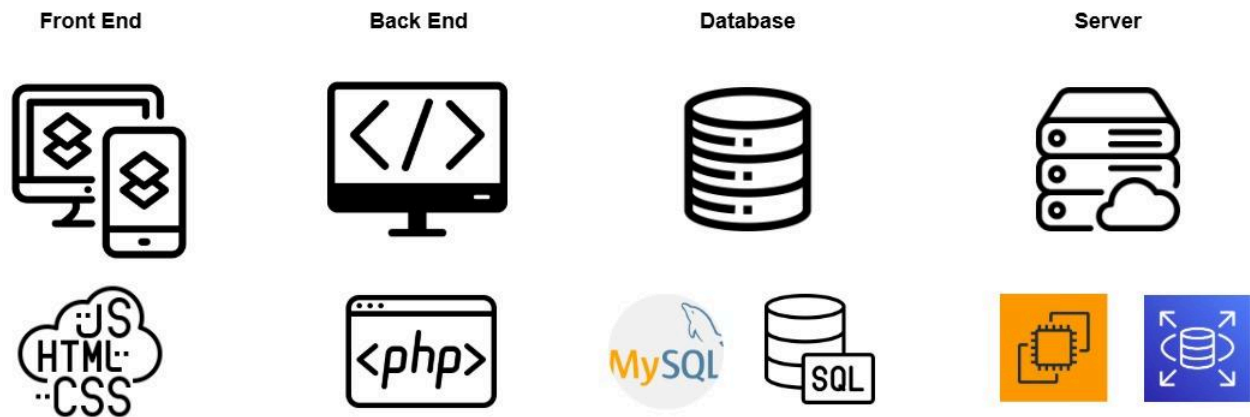
Save profile

Username is fixed for login; wire MySQL later to allow admin-approved changes.

3. Technical Requirements

3.1 Project Architecture

Put a diagram how you would show your tech-stack.



Include a diagram and a brief description of each tech stack used.

Front End

HTML: Provides the basic structure of web pages.

CSS: Handles styling, layout, and responsiveness.

JavaScript (JS): Adds interactivity and dynamic behavior on the client side.

Back End

PHP: A server-side scripting language used to process requests, handle business logic, manage sessions, and communicate with databases.

Database

MySQL / SQL: Relational database system used to store, retrieve, and manage structured data using SQL (Structured Query Language).

Server

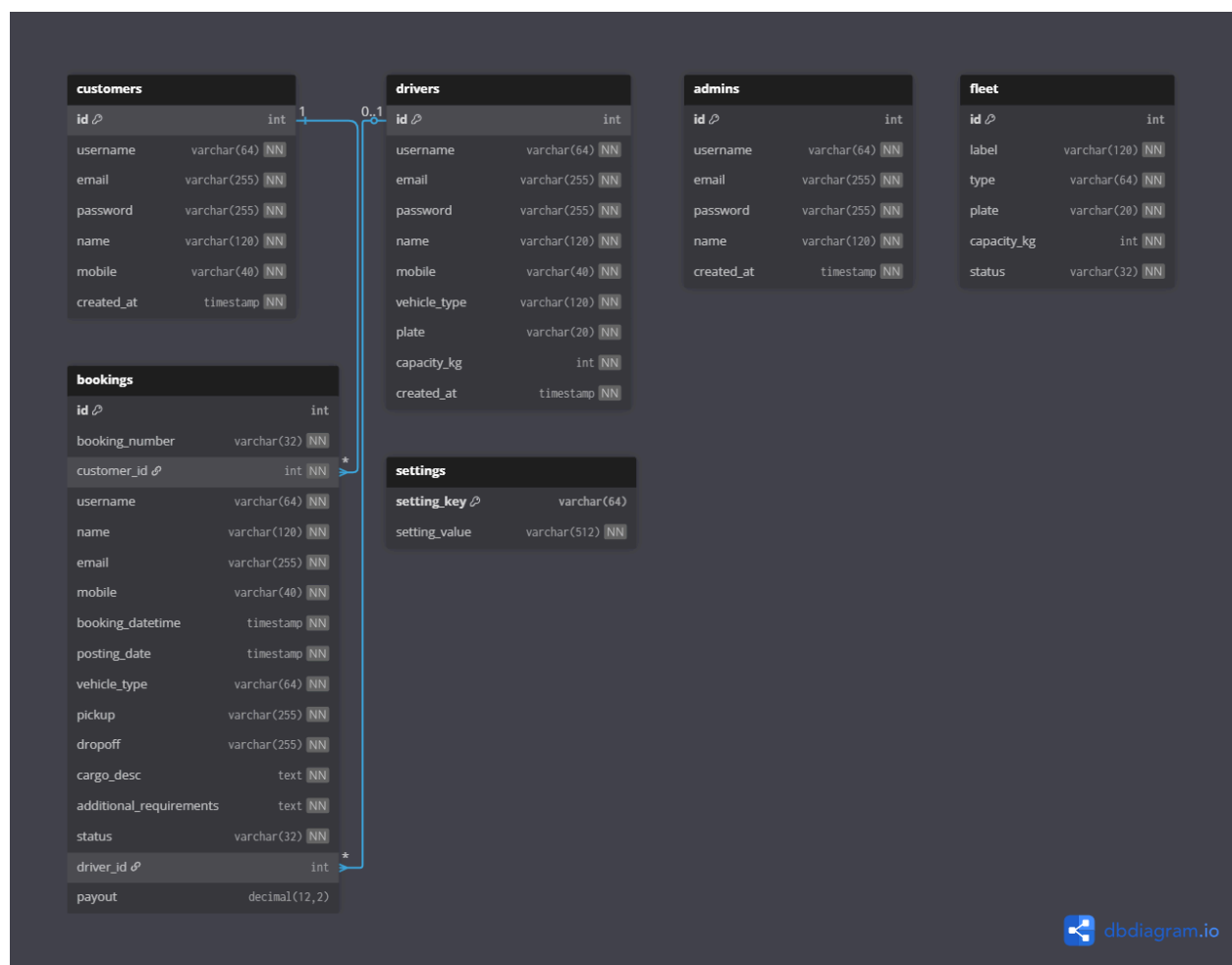
Amazon EC2: Provides scalable virtual machines to run your back-end code. It handles compute power, operating system, and deployment environment.

Amazon RDS: A managed service for running databases like MySQL without handling manual setup, backups, or maintenance.

Amazon Aurora: A cloud-optimized database compatible with MySQL, offering better performance and scalability than standard RDS MySQL.

7.1. Database Design

Add a data dictionary for your database here.



The image presents a **database structure (ERD)** for a booking or delivery system, showing how different pieces of information are connected.

The main parts of the system are **customers, drivers, and bookings**. A customer can make many bookings, but each booking belongs to only one customer. Likewise, a driver can handle multiple bookings, but each booking is assigned to just one driver. These connections show how data flows between users and services.

The **bookings table** is the most important part because it contains all the transaction details, such as pickup and drop-off locations, schedule, vehicle type, and booking status. It links both the customer and the driver, making it the center of the system.

Other tables support the system:

- **Admin** stores information about the booking management system of GK services.
- The **fleet** contains details about vehicles.
- **Settings** keep system configuration values.

Overall, the diagram shows how the system organizes and connects data to manage bookings, users, and vehicles efficiently.

Complementing the Entity Relationship Diagram (ERD), a visual representation of the system's components and how they interact with each other are shown with the use of Architecture Diagram.

Customer Panel: Grants customers to such privileges: booking of trucks, tracking of shipments, list of bookings and shipments, and profile setup, allowing an interactive engagement with GK services.

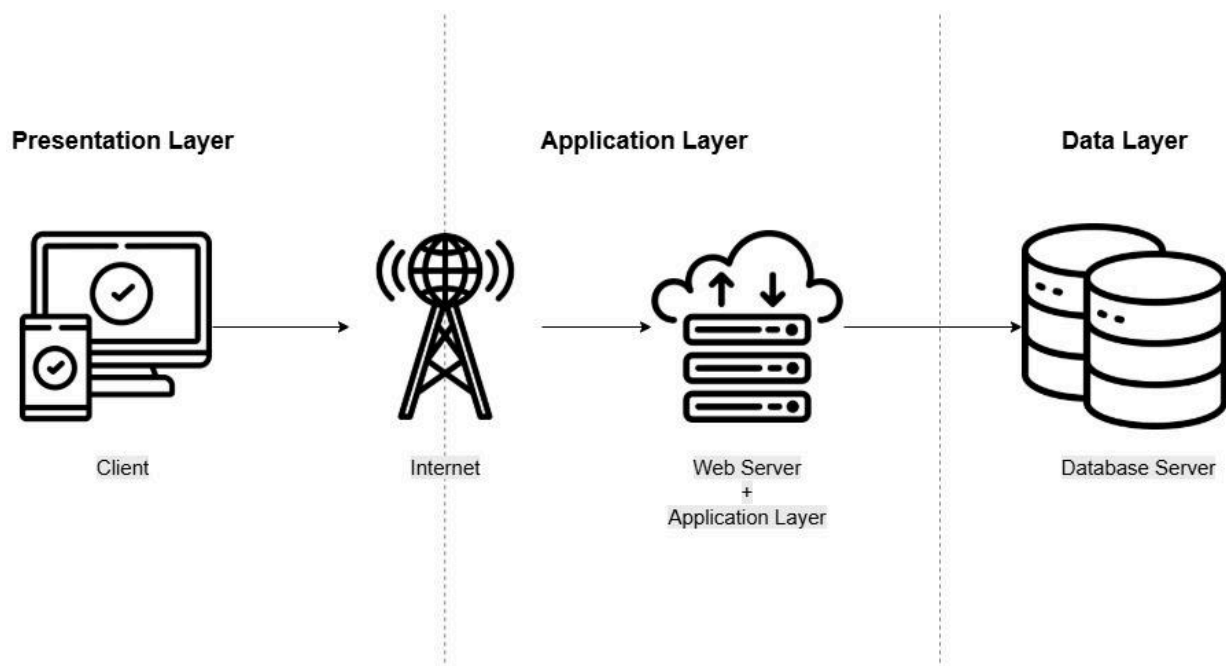
Admin Panel: Provides the following access to the operation controls of GK services: list of bookings (managing status of bookings and uploading of gate pass), driver management, customer management, and fleet management to ensure the system runs effectively.

Driver Panel: Allows drivers to view available bookings (accepting jobs after admin approval), list of deliveries, and earnings that will reflect on their dashboard after the booking is delivered.

The PHP backend serves as the centralized application layer, facilitating communication between both interfaces and the MySQL database, where all data defined in the Entity-Relationship Diagram (ERD) is securely maintained and managed.

7.2. Cloud Requirements

Put here a diagram of your cloud architecture as well as a short description.



Presentation Layer (Client + Internet)

This is the **entry point** for users.

- **Client (Browser / Mobile App):**
Built with HTML, CSS, and JavaScript, this is where users interact with our system: viewing pages, submitting forms, and triggering requests.
 - **Internet / Networking Layer:**
Requests travel over the internet using HTTP. In a cloud setup, this is typically routed through services like DNS and load balancers before reaching your application.
-

Application Layer (Web Server + Business Logic)

This is the **core processing layer** where your system's logic lives.

- **Web Server:**
Handles incoming HTTP requests and serves responses (HTML, JSON, etc.).
 - **Application Logic:**
Processes user input, enforces business rules, manages authentication, and interacts with the database.
 - **Cloud Mapping:**
 - **Amazon EC2** runs your web server and PHP application.
 - Often paired with **Auto Scaling** and **Elastic Load Balancing** for reliability and performance.
-

Data Layer (Database Server)

This is where persistent data is stored and managed.

- **Relational Database (MySQL/SQL):**
Stores structured data such as user accounts, transactions, and application records.
- **Cloud Mapping:**
 - **Amazon RDS** provides managed MySQL databases.
 - **Amazon Aurora** offers enhanced performance, scalability, and availability.

4. Appendix

4.1 Deployment Instructions

This section provides a step-by-step guide for deploying the GRK Trucking Services Web-Based Booking and Tracking Management System using a cloud environment.

Step 1: Prepare Local Project Files

1. Ensure all project files are complete and working locally:
 - Front-end: HTML, CSS, JavaScript
 - Back-end: PHP
 - Database: MySQL
2. Test the system using XAMPP or any local server to verify:
 - Login and registration are functioning
 - Booking system works correctly
 - Database connections are stable

Step 2: Set Up Amazon EC2 Instance

1. Log in to your AWS account.
2. Navigate to EC2 Dashboard.
3. Click **Launch Instance**.
4. Configure the instance:
 - Choose **Amazon Linux or Ubuntu**
 - Select instance type (e.g., t2.micro for free tier)
5. Create or select an existing key pair.
6. Configure security group:
 - Allow **HTTP (Port 80)**
 - Allow **SSH (Port 22)**
7. Launch the instance.

Step 3: Connect to EC2 Server

1. Open terminal or PuTTY.
2. Connect using your key pair:
`ssh -i your-key.pem ec2-user@your-public-ip`

Step 4: Install Required Software

Inside the EC2 instance, install Apache, PHP, and MySQL client:

For Ubuntu:

```
sudo apt update
```

```
sudo apt install apache2 php mysql-client -y
```

For Amazon Linux:

```
sudo yum update -y
```

```
sudo yum install httpd php mysql -y
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

Step 5: Upload Project Files

1. Use SCP, FileZilla, or WinSCP to transfer files:

```
scp -i your-key.pem -r project-folder ec2-user@your-ip:/var/www/html/
```

2. Move files to the web directory if needed:

```
sudo mv project-folder/* /var/www/html/
```

Step 6: Configure File Permissions

```
sudo chmod -R 755 /var/www/html/
```

Step 7: Set Up Database (Amazon RDS)

1. Go to AWS RDS Dashboard.
2. Create a new MySQL database instance.
3. Configure:
 - Database name
 - Username and password
4. Allow public access or connect EC2 security group.

Step 8: Import Database

1. Export your local database (from XAMPP/phpMyAdmin).
2. Import it to RDS:

```
mysql -h your-rds-endpoint -u admin -p database_name < database.sql
```

Step 9: Configure Database Connection

1. Open your PHP config file (e.g., `config.php`).
2. Update database credentials:

```
$host = "your-rds-endpoint";
```

```
$user = "username";
```

```
$pass = "password";
```

```
$db = "database_name";
```

Step 10: Restart Server

```
sudo systemctl restart apache2
```

or

```
sudo systemctl restart httpd
```

Step 11: Access the Application

1. Open a browser.
2. Enter your EC2 public IP:
`http://your-public-ip`
3. Verify that:
 - Pages load correctly
 - CSS and design appear properly
 - System functions (login, booking, tracking) are working

Step 12: Final Testing and Optimization

- Test all user roles (Admin, Driver, Customer)
- Check real-time booking and tracking
- Fix any issues related to file paths or CSS loading
- Optionally configure:
 - Domain name (via Route 53)
 - SSL certificate (HTTPS)

This deployment process ensures that the system is fully accessible online, scalable, and ready for real-world use through cloud infrastructure.

4.2 Testimonials

Student: Shaina Mae Villasis

Instructor: Joseph Domingo

Date: May 2026

For this semester, our project focused on developing a logistics system called GRK Trucking Services, similar to Lalamove. This project provided a new and challenging experience compared to the systems I usually develop. As we defined the objectives and requirements, I realized that building a logistics system is complex, as it involves multiple processes, careful planning, and detailed system design.

During the design phase, I mapped out the business architecture and divided it into manageable modules. Throughout the development process, I did not encounter significant issues with either the front-end or back-end. However, challenges arose during the deployment stage using AWS EC2. Although the system was successfully deployed, the CSS did not load properly on the live website. After identifying that the issue was related to the structure of our modules, I resolved it by correctly integrating the HTML and CSS, ensuring that the design displayed properly in the live environment.

In the final three weeks of development, we were required to use RDS (MySQL) as our database. Initially, our team had limited knowledge of this technology, but we adapted quickly with the guidance of our instructor, Sir Pat. Since we were already using XAMPP MySQL for our local database, the transition to RDS was manageable. We exported our local database and migrated it to RDS with minimal configuration.

During the testing phase, we verified that all system panels were functioning correctly. In addition, I implemented access control measures to ensure that users could only access the features and panels authorized for them.

Through this experience, I learned that cloud deployment is not merely a technical requirement but a critical component of system development. It significantly impacts system performance, user experience, and scalability. Regardless of how well-developed a system is, it loses its value if it is not accessible or reliable. This project enhanced my understanding of deploying and managing systems in real-world environments, including handling user access and ensuring system stability.

Student: Christine Jane Villasis

Instructor: Joseph Domingo

Date: May 2026

This course highlighted that the backbone of any successful information system isn't just the code, but the research and analytical planning that define it. My role focused on the initial stages of the project, where I gathered and synthesized the critical data necessary to build our foundation—specifically for the Background of the Study, Objectives, and Scope.

I began by analyzing the Lalamove ecosystem and current logistics market trends to identify the specific manual bottlenecks within GRK Trucking Services. This research was essential for drafting a compelling Background of the Study that justified our transition to a digital solution. Furthermore, creating the Objectives and Scope taught me how to set realistic, data-driven boundaries for our B2B operations and AWS deployment. This process of turning raw information into a structured project roadmap allowed me to see how technical requirements must directly align with business needs.

In conclusion, these tasks significantly sharpened my ability to interpret data and document complex processes. This experience is directly relevant to my future goal of becoming a Business Analyst or Data Analyst, as I now feel confident in my ability to research industry-wide problems and transform raw information into clear, actionable project requirements.”

Student: Alyzza Marizz Esmeña

Instructor: Joseph Domingo

Date: May 2026

This course gave me hands-on and new experience in documentary analysis, cloud-based system documentation, and freight logistics research. I was responsible for searching and synthesizing the Review of Related Literature (RRL), defining all features for the GRK Trucking Services platform, and writing the summary and conclusion.

I learned how to translate real-world trucking needs—such as batch and container-scale freight management—into clear technical requirements, including the

Landing Page, Admin Dashboard, and Driver Dashboard. Using Google Docs and cloud architecture tools, I documented every stage of the project.

This knowledge directly supports my goal of working as a systems analyst or technical writer in logistics or cloud computing. I highly recommend this course to any student interested in understanding how research and documentation drive successful real-world applications.

Student: Jhay Cyrus J. Angeles

Instructor: Joseph Domingo

Date: May 2026

A significant part of the final development phase involved transitioning our database to AWS RDS using MySQL. Initially, our team had limited experience with RDS and MySQL Workbench, but with guidance from our instructor, Sir Pat, we quickly adapted. Since we were already using XAMPP MySQL locally, we exported our database and connected it to RDS through MySQL Workbench.

We then configured the necessary config.php and info.php files to align with the connection requirements of the RDS environment, ensuring proper database access and secure communication between the application and the cloud database. During testing, we verified that all system panels functioned correctly and that the database connections were stable.

Through this experience, I gained a deeper understanding of how critical proper database configuration and cloud integration are in system development. Connecting the system to RDS and configuring the required files was not just a technical step, but an essential process that directly impacted performance, reliability, and scalability. This project strengthened my ability to deploy and manage systems in a real-world cloud environment while ensuring secure and efficient database connectivity.